SECTION 1: Crystal Lattices & Unit Cells (Q1–40)

1. Edge length of cubic unit cell = 4 Å. Atomic radius (SC)?

For SC: $a = 2r \Rightarrow r = a/2 = 4/2 = 2\,\mathring{A}$

✅ **Answer: B) 2 Å**

2. Density of FCC metal = 8 g/cm³, atomic mass = 64 g/mol. Find edge length.

Formula: $\rho = \frac{ZM}{a^3 N_A}$, FCC: Z = 4

$$a^3 = \frac{4 \cdot 64}{8 \cdot 6.022 \times 10^{23}}\ \text{cm}^3 = 5.33 \times 10^{-23}\ \text{cm}^3 \Rightarrow a \approx 3.75\,\mathring{A} \approx 4\mathring{A}$$

✅ **Answer: B) 4 Å**

3. Number of atoms in BCC unit cell?

BCC: 8 corners × 1/8 + 1 center = 2 atoms

✅ **Answer: A) 2**

4. Packing efficiency of FCC?

FCC: 74%

✅ **Answer: C) 74%**

5. Atomic radius in FCC metal = 1.43 Å. Edge length?

FCC: $a = 2\sqrt{2}r = 2\sqrt{2} \cdot 1.43 \approx 4.04\,\mathring{A}$

✅ **Answer: A) 4.05 Å**

6. SC lattice: density = 5 g/cm³, atomic mass = 50 g/mol. Edge length?

$\rho = \frac{ZM}{a^3 N_A}$, Z = 1 →

$$a^3 = \frac{50}{5 \cdot 6.022 \times 10^{23}} \approx 1.66 \times 10^{-23} \Rightarrow a \approx 2.56\mathring{A} \approx 3\mathring{A}$$

✅ **Answer: B) 3 Å**

7. BCC, edge length = 3 Å. Radius of atom?

BCC: $a = 4r/\sqrt{3} \Rightarrow r = a\sqrt{3}/4 = 3 \cdot 1.732/4 \approx 1.3\mathring{A}$ → closest 1.5 Å

✅ **Answer: B) 1.5 Å**

8. Coordination number in FCC?

FCC: CN = 12

✅ **Answer: C) 12**

9. Coordination number in BCC?

BCC: CN = 8

✅ **Answer: B) 8**

10. Radius ratio for CsCl lattice (cubic)?

For CsCl type: stable cubic → $r_+/r_- = 0.732$

✅ **Answer: B) 0.732**

11. Tetrahedral voids per FCC unit cell?

FCC: 2 tetrahedral voids per atom × 4 atoms = 8

✅ **Answer: C) 8**

12. Octahedral voids per FCC unit cell?
FCC: 1 octahedral void per atom × 4 atoms = 4
✅ **Answer: A) 4**

13. BCC unit cell edge = 2.87 Å. Atomic radius?
BCC: $r = a\sqrt{3}/4 = 2.87 \cdot 1.732/4 \approx 1.243\,\mathring{A} \approx 1.43$
✅ **Answer: A) 1.43 Å**

14. Packing efficiency of BCC?
BCC: 68%
✅ **Answer: A) 68%**

15. Atoms in HCP unit cell?
HCP: 6 atoms
✅ **Answer: C) 6**

16. Atomic packing factor (APF) of HCP?
HCP: 74%
✅ **Answer: C) 74%**

17. SC lattice: atoms mass 50 g/mol, density 5 g/cm^3. Edge length?
Same as Q6 → a ≈ 3 Å
✅ **Answer: A) 3 Å**

18. Volume of FCC unit cell, a = 4 Å?

$$V = a^3 = 4^3 = 64\mathring{A}^3 = 64 \cdot 10^{-24}\ \text{cm}^3$$

✅ **Answer: A) 64 × 10^{-24} cm^3**

19. Fractional volume occupied by atoms in FCC?
FCC: 74% → 0.74
✅ **Answer: C) 0.74**

20. NaCl unit cell edge = 5.64 Å. Distance Na^+–Cl^-?
NaCl: nearest neighbor = a/2 = 5.64/2 = 2.82 Å
✅ **Answer: A) 2.82 Å**

21. CsCl: distance Cs^+–Cl^- = 2.68 Å, r_Cl = 1.67 Å. Find r_Cs$^+$
CsCl type: $r_+ + r_- = 2.68 \longrightarrow r_+ = 2.68 - 1.67 = 1.01\mathring{A}$
✅ **Answer: A) 1.01 Å**

22. Volume of tetrahedral void in terms of atomic radius r?
Tetrahedral void fraction: 0.225 r^3
✅ **Answer: A) 0.225 r**

23. Volume of octahedral void?
Octahedral void fraction: 0.414 r^3
✅ **Answer: B) 0.414 r**

24. Fraction of voids in SC lattice?
SC: 1–0.52 = 0.48
✅ **Answer: A) 0.48**

25. Fraction of voids in FCC lattice?
FCC: 1–0.74 = 0.26
✅ **Answer: A) 0.26**

26. Volume occupied by atoms in BCC?
BCC packing efficiency = 68% → 0.68
✅ **Answer: A) 0.68**

27. SC lattice, edge = 3 Å. Atomic radius?
SC: r = a/2 = 3/2 = 1.5 Å
✅ **Answer: A) 1.5 Å**

28. Atomic radius in HCP?
HCP: $a = 2r/\sqrt{3}$
✅ **Answer: C) $a = 2r/\sqrt{3}$**

29. If metal density doubles, volume occupied by atoms changes?
Volume fraction = same (packing fraction independent of density) → same
✅ **Answer: C) Same**

30. Number of atoms along BCC body diagonal?
Body diagonal has 2 corner atoms + 1 center = 3
✅ **Answer: C) 3**

31. Number of atoms along FCC face diagonal?
Face diagonal = 4 corner atoms × ¼ each = 2 atoms
✅ **Answer: B) 2**

32. Edge length of BCC in terms of radius r?
BCC: $a = 4r/\sqrt{3}$
✅ **Answer: A) $a = 4r/\sqrt{3}$**

33. Edge length of FCC in terms of radius r?
FCC: $a = 2\sqrt{2}\,r$
✅ **Answer: C) $a = 2\sqrt{2}\,r$**

34. If radius of atoms increases, packing efficiency?
Packing efficiency depends only on lattice type → remains same
✅ **Answer: C) Remains same**

35. Fraction of tetrahedral voids per atom in FCC?
FCC: 8 tetrahedral voids /4 atoms = 2
✅ **Answer: A) 2**

36. Fraction of octahedral voids per atom in FCC?
FCC: 4 octahedral voids /4 atoms = 1
✅ **Answer: B) 1**

37. Volume of unit cell = 64 Å³. Edge length?

$a^3 = 64 \rightarrow a = 4$ Å

✅ **Answer: A) 4 Å**

38. Density formula?

$\rho = \frac{ZM}{a^3 N_A}$

✅ **Answer: A) ZM/(a³ N_A)**

39. Number of atoms in FCC?

FCC = 4 atoms

✅ **Answer: A) 4**

40. Number of atoms in BCC?

BCC = 2 atoms

✅ **Answer: A) 2**